

Virtual Draw of Arbitrary Geometry Hollow Core Fibres, a 3D Finite Element Approach.

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Abstract A novel 3D finite-element Computational Fluid Design tool is developed to study the fabrication of anti-resonant hollow-core fibres from preforms with arbitrary geometries. To showcase its unique capabilities, we study the draw of elliptical tubes and of hollow-core fibres with elliptical capillaries. @2026 The Author(s)

Introduction

Hollow-core optical fibres (HCFs) have attracted significant attention because they guide light mainly in air, reducing nonlinear effects, material absorption, and latency compared with conventional solid-core fibres. These advantages make HCFs attractive for high-power transmission, ultrafast optics, and low-loss signal propagation [1].

Performance of hollow-core fibres is strongly governed by microstructure geometry, and recent studies have shown that some designs can offer clear optical advantages over conventional structures consisting of assemblies of circular tubes [2, 3]. For instance, use of elliptical tubes can reduce confinement loss, improve single-mode guidance, and offer greater flexibility in tailoring dispersion and confinement properties [4]. However, so far no elliptical tube HCF has been realised in practice.

Several analytical and numerical approaches have been developed to study optical fibre fabrication [5-7]. However, most existing models were developed for circular or simplified geometries and cannot directly support arbitrary cross-sections [5, 7]. Although some methods have been developed for more arbitrary or asymmetric structures [8], none of them has been applied to HCF.

In this work, a new 3D finite-element computational fluid dynamic (CFD) method is developed in ANSYS Polyflow 2025 as a first step toward the simulation of arbitrary-geometry fibre drawing. This approach is first validated against an experimental tube draw. Then it is applied to an elliptical jacket tube to examine the effects of draw parameters on ellipticity preservation during the draw. It is then applied to a tubular anti-resonant hollow-core fibre containing both circular and elliptical capillaries. This provides a foundation for studying the fabrication of optical fibres with arbitrary geometries and offers a practical route toward the design and manufacture of advanced hollow-core fibre structures.

Modelling approach

In general, the fibre fabrication process is modelled as a steady, laminar flow of an incompressible Newtonian fluid governed by the Navier–Stokes equations [9] :

$$\frac{\partial v_k}{\partial x_k} = 0, \rho \left(v_i \frac{\partial v_j}{\partial x_i} \right) = -\frac{\partial p}{\partial x_j} + \frac{\partial \tau_{ij}}{\partial x_i} + \rho g_j \quad (1)$$

where v_k denotes the velocity components, x_k the spatial coordinates, and p the pressure. The density is taken as constant, $\rho = 2200 \text{ kg/m}^3$. For a Newtonian material, the extra stress tensor is

$$\tau_{ij} = \mu \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j} \right) - \frac{2}{3} \mu \delta_{ij} \frac{\partial v_k}{\partial x_k} \quad (2)$$

where μ is the dynamic viscosity. The glass temperature was calculated using Eq. (3), where $T(z)$ is the glass temperature at the axial position z , $T_{\max}$ (K) is the maximum glass temperature. In the present study, z_c , β , and q were kept constant to $z_c = 0.046 \text{ (m)}$, $\beta = 0.058 \text{ (m)}$, and $q = 2.76$. $T_{\max}$ is considered as a free-parameter and adjusted to modify the draw tension.

$$T(z) = T_{\max} \exp \left[-\frac{1}{2} \left| \frac{z - z_c}{\beta} \right|^q \right] \quad (3)$$

The material being drawn is pure fused silica with surface tension of $\gamma = 0.3 \text{ N/m}$, and the viscosity is strongly temperature-dependent, following an Arrhenius-type relation [5] :

$$\mu(T) = 0.58 \times 10^{-7} \times \exp \left(\frac{515400}{8.314T} \right) \quad (4)$$

The eccentricity of ellipse is defined as:

$$e = (1 - b^2/a^2)^{1/2} \quad (5)$$

where a and b are the major and minor axes of the ellipse, respectively. For the numerical simulations, a streamwise remeshing technique and an optimized evolution strategy, with case-specific incremental steps chosen according to the requirements of each simulation, are implemented to address the large aspect ratio of the computational domain and to impose the boundary conditions progressively. These developments were essential to enable the software to

carry out the simulations robustly and reliably; without them, the software itself was unable to perform the simulations. Also, a quarter-wedge symmetric model was employed, using a 16,000-element mesh comprising 400 nodes along the axial direction, 20 in the radial direction, and 2 through the thickness, as determined by a grid-independence study.

Validation

For validation, 3D simulations of a circular capillary draw were compared with experimental and numerical data from the literature [5]. The initial preform, with outer and inner radii of 9.96 mm and 3.43 mm, was drawn to a fibre with radii of 180 μm and 90 μm , respectively (50% expansion ratio). Simulations were performed for seven draw tension values. For each case, the internal pressure required to maintain a constant expansion ratio of 50% was determined and plotted in Fig. 1. The predicted draw tension–internal pressure relationship agrees closely with experimental results over the full range and shows better agreement than the previous numerical study [5]. The maximum deviation at higher pressures is below 4%, demonstrating the suitability of this numerical method.

Modelling an elliptical jacket tube

To investigate how non-circular geometries evolve during the draw process, an elliptical tube preform with an initial eccentricity (e_0) of 0.8 was considered, using the same cross-sectional area as in the validation study above [5]. The simulation was performed at a draw tension of 50 g, an applied gas pressure in the preform of 1.4 kPa, and a draw ratio (the ratio of the draw speed to the feed speed) of 2000. The evolution of the major and minor axes and jacket eccentricity through the neckdown is shown in Fig. 2. The major axis decreases continuously, whereas the minor axis initially increases, reaches a maximum, and then decreases. This indicates a continuous reduction in eccentricity, and the final cross-section becomes nearly circular.

This reduction in ellipticity results from the interplay of viscous stresses, surface tension, and internal gas pressure in the hot zone. As temperature rises, viscosity decreases significantly, reducing viscous resistance and making the balance mainly dependent on surface tension and internal pressure. Because the local radius of curvature is smaller along the major axis, surface tension is stronger in that direction. Consequently, the internal pressure cannot overcome the combined effects of surface tension and viscous stresses, and the major axis decreases. In contrast, the larger radius of curvature along the

minor axis reduces surface tension, allowing internal pressure to initially dominate and increase the minor axis. This reduces the difference between the effective radii of the two axes; the remaining difference leads to unequal surface tension along each axis, which progressively drives the eccentricity to zero, resulting in a circular shape.

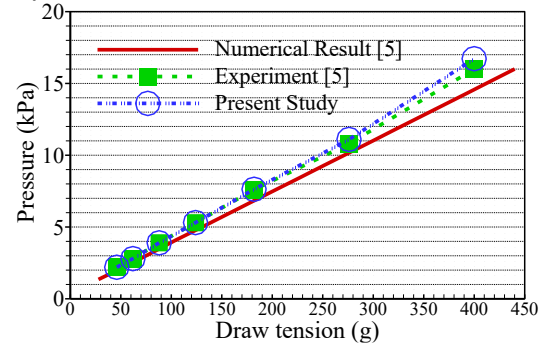


Fig. 1: Validation of internal pressure versus draw tension for the present study and previous studies.

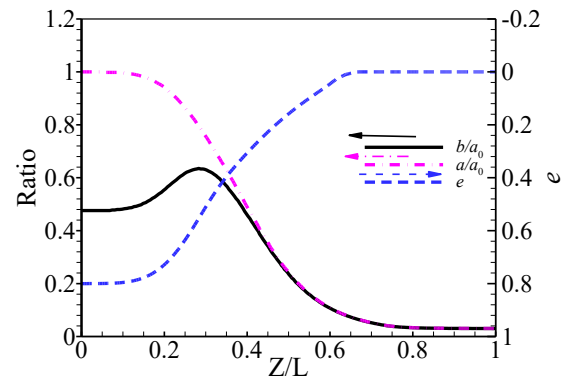


Fig. 2: Evolution of the major (a), minor (b) axes and eccentricity along the calculation length (L).

Effect of draw tension and gas pressure on ellipticity preservation of jacket tube

Figure 3 (a) shows the eccentricity of a jacket tube drawn from an initial jacket tube preform with $e_0 = 0.8$ at a draw ratio of 2000 under different draw tensions and internal gas pressures. At low draw tension, lower viscosity in the hot zone reduces viscous stresses, allowing surface-tension-driven relaxation to reduce eccentricity even without internal pressure, so the fibre cross section becomes more circular. As draw tension increases, viscous stresses become stronger and better preserve the initial elliptical geometry.

For a fixed draw tension, increasing internal gas pressure causes a further decrease in eccentricity because the pressure more effectively counteracts the combined effects of surface tension and viscous stresses, especially along the minor axis. Thus, preserving an elliptical cross section is favoured by high draw tension and low internal gas pressure.

Figure 3(b) shows the variation of eccentricity

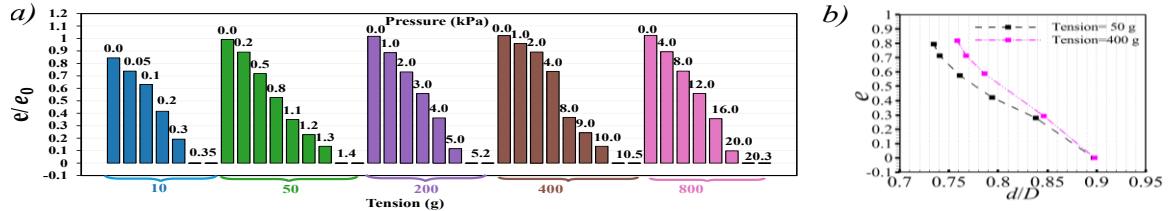


Fig. 3: Effect of internal gas pressure and draw tension on the final fibre eccentricity: **(a)** eccentricity evolution under varying internal gas pressures and draw tensions, **(b)** dependence of eccentricity on diameter ratio (d/D).

with the d/D ratio of the fibre fabricated from the selected preform with an initial major-axis d/D of 0.767. As the applied pressure increases, this ratio changes within the fabricated jacket tube. Here, d/D is defined as the ratio of the inner to outer major diameter. Applying pressure increases the ratio d/D , but also reduces the eccentricity. And as expected, higher tension can help preserve the eccentricity.

Modelling approach for anti-resonant hollow core fibres

To demonstrate the applicability of the developed method to a different and technologically relevant fibre geometry, a tubular anti-resonant hollow-core fibre was considered. Based on an experimental study [5], the preform properties of the selected six-capillary configuration (6C) were taken as follows: a cane capillary diameter ratio of $d_c/D_c=0.67$, a cane capillary outer diameter of $440\text{ }\mu\text{m}$, a cane diameter of 2.9 mm , and jacket tube dimensions of $d/D=3/10\text{ mm}$. The corresponding draw conditions were a draw tension of 300 g , a draw stress of 98 MPa , and a fibre yield of 2.4 km (per metre of preform). For the present simulation, a capillary differential pressure of 12 kPa was considered.

As shown in Fig. 4, two different preform geometries were considered, as illustrated in Fig. 4(a) and Fig. 4(b). In both cases, the capillaries were placed in the same overall arrangement, and it was assumed that half of the capillary wall thickness was embedded within the cane at the preform stage. The geometry in Fig. 4(a) corresponds to a preform with circular capillaries, whereas in Fig. 4(b) the capillaries were taken to be elliptical while preserving the same capillary cross-sectional area. For the elliptical case, the capillary shape in the preform was defined by the ratio $b_{c,p}/a_{c,p}=0.6$, where $a_{c,p}$ and $b_{c,p}$ are the major and minor axes of the capillary in the preform, respectively. This corresponds to a capillary eccentricity of $e=0.8$. The two preforms therefore differed only in the initial capillary shape and were numerically drawn with our 3D model under the following drawing conditions: a draw tension of 300 g , a draw stress of 98 MPa , and a fibre yield of 2.4 km per metre of preform. In both cases, a differential pressure between core and capillary of 12 kPa was considered.

The resulting capillary shapes in the final fibre were then compared using the ratio $b_{c,f}/a_{c,f}$, where

$a_{c,f}$ and $b_{c,f}$ denote the major and minor axes of the capillary in the fibre, respectively. For the case shown in Fig. 4(a), the final capillary axis ratio was found to be $b_{c,f}/a_{c,f}=1.08$, caused by the initial capillary indentation into the cane, while for the case shown in Fig. 4(b) it was $b_{c,f}/a_{c,f}=1.04$. These results show that introducing eccentricity into the capillary at the preform stage did not result in the formation of an elliptical capillary in the final fibre. Instead, it only caused a slight change in the final capillary axis ratio. This suggests that prescribing capillary eccentricity in the preform alone is not sufficient to produce an elliptical capillary after drawing. Rather, the final capillary geometry is expected to depend on a broader combination of geometric and processing parameters, the effects of which will be investigated in future work.

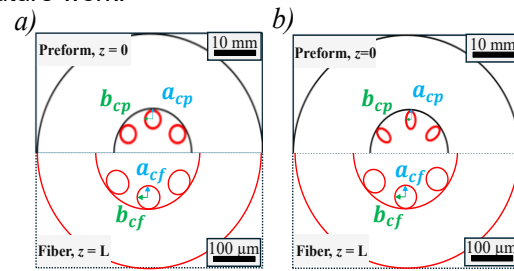


Fig. 4: **(a)** Preform and final fibre geometry for the circular-capillary case, and **(b)** preform and final fibre geometry for the elliptical-capillary case.

Conclusion

We used commercial CFD software to model HCF fibre drawing with arbitrary shaped tubes. The developed 3D finite-element CFD model showed that elliptical jacket tubes can be successfully drawn under certain conditions, with ellipticity better preserved at higher draw tension and lower internal pressure. To achieve a larger eccentricity in the final jacket, a greater initial preform eccentricity and a larger initial preform area was found to be more favourable. When applied to elliptical microstructure of a six-capillary anti-resonant hollow-core fibre, numerical modelling shows that introducing capillary eccentricity in the preform alone does not produce an elliptical final capillary, but only slightly altered the final shape. These results demonstrate that 3D CFD modelling can be used to study the drawing process for HCFs and could form the basis for future studies exploring the production of hollow-core fibres with more complex geometries beyond just ellipticity.

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